6. Develop a C program for a Doubly Linked List representing a sequence of web pages visited by a user. The program should insert a new page, move forward and backward, delete a specified page, and display the pages from first-to-last and last-to-first while handling beginning and end conditions correctly.

Source code :-

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

struct Node

{

   char page[50];

   struct Node *prev;

   struct Node *next;

};

struct Node *head = NULL;

struct Node *current = NULL;

void insertPage()

{

   char page[50];

   struct Node *newNode;

   printf("Enter page name: ");

   scanf("%s", page);

   newNode = malloc(sizeof(struct Node));

   strcpy(newNode->page, page);

   newNode->prev = NULL;

   newNode->next = NULL;

   if (head == NULL)

   {

      head = newNode;

      current = newNode;

   }

   else
```

```
    {
        current->next = newNode;
        newNode->prev = current;
        current = newNode;
    }
    printf("Page inserted\n");
}
void moveForward()
{
    if (current == NULL)
        printf("No pages available\n");
    else if (current->next == NULL)
        printf("Already at the last page\n");
    else
    {
        current = current->next;
        printf("Current page: %s\n", current->page);
    }
}
void moveBackward()
{
    if (current == NULL)
        printf("No pages available\n");
    else if (current->prev == NULL)
        printf("Already at the first page\n");
    else
    {
        current = current->prev;
        printf("Current page: %s\n", current->page);
    }
}
```

```
void deletePage()
{
   char page[50];
   struct Node *temp;
   printf("Enter page to delete: ");
   scanf("%s", page);
   temp = head;
   while (temp != NULL && strcmp(temp->page, page) != 0)
      temp = temp->next;
   if (temp == NULL)
   {
      printf("Page not found\n");
      return;
   }
   if (temp->prev != NULL)
      temp->prev->next = temp->next;
   else
      head = temp->next;
   if (temp->next != NULL)
      temp->next->prev = temp->prev;
   if (current == temp)
   {
      if (temp->next != NULL)
         current = temp->next;
      else
         current = temp->prev;
   }
   free(temp);
   printf("Page deleted\n");
}
```

```
void displayForward()
{
   struct Node *temp = head;
   if (head == NULL)
   {
      printf("No pages available\n");
      return;
   }
   printf("Pages (First to Last): ");
   while (temp != NULL)
   {
      printf("%s <-> ", temp->page);
      temp = temp->next;
   }
   printf("NULL\n");
}
void displayBackward()
{
   struct Node *temp = head;
   if (head == NULL)
   {
      printf("No pages available\n");
      return;
   }
   while (temp->next != NULL)
      temp = temp->next;
   printf("Pages (Last to First): ");
   while (temp != NULL)
   {
      printf("%s <-> ", temp->page);
```

```
        temp = temp->prev;
    }
    printf("NULL\n");
}
int main()
{
    int choice;
    while (1)
    {
        printf("\n--- Web Page History ---\n");
        printf("1. Insert Page\n");
        printf("2. Move Forward\n");
        printf("3. Move Backward\n");
        printf("4. Delete Page\n");
        printf("5. Display First to Last\n");
        printf("6. Display Last to First\n");
        printf("7. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
                insertPage();
                break;
            case 2:
                moveForward();
                break;
            case 3:
                moveBackward();
                break;
            case 4:
```

```
            deletePage();
            break;
        case 5:
            displayForward();
            break;
        case 6:
            displayBackward();
            break;
        case 7:
            return 0;
        default:
            printf("Invalid choice\n");
      }
   }
   return 0;
}
```

Output :-

```
--- Web Page History ---
1. Insert Page
2. Move Forward
3. Move Backward
4. Delete Page
5. Display First to Last
6. Display Last to First
7. Exit
Enter choice: 1
Enter page name: aswanth
Page inserted

--- Web Page History ---
1. Insert Page
2. Move Forward
3. Move Backward
4. Delete Page
5. Display First to Last
6. Display Last to First
7. Exit
Enter choice: 1
Enter page name: sai
Page inserted

--- Web Page History ---
1. Insert Page
2. Move Forward
3. Move Backward
4. Delete Page
5. Display First to Last
6. Display Last to First
7. Exit
Enter choice: 5
Pages (First to Last): aswanth <-> sai <-> NULL
```